Supplemental Material

Bottom-up impacts of forecasted climate change on the eastern Bering Sea food web

George A. Whitehouse[1,2,3]*, Kerim Y. Aydin[2], Anne B. Hollowed[2], Kirstin K. Holsman[2], Wei Cheng[1,4], Amanda Faig[3], Alan C. Haynie[2], Albert J. Hermann[1,4], Kelly A. Kearney[1,2], André E. Punt[3], and Timothy E. Essington[3]

[1] Joint Institute for the Study of the Atmosphere and Ocean, now the Cooperative Institute for Climate, Ocean, and Ecosystem Studies, University of Washington, Seattle, WA, United States
[2] Alaska Fisheries Science Center, National Marine Fisheries Service, National Oceanic and Atmospheric Administration, Seattle, WA, United States
[3] School of Aquatic and Fishery Sciences, University of Washington, Seattle, WA, United States
[4] Pacific Marine Environmental Laboratory, Oceans and Atmospheric Research, National Oceanic and Atmospheric Administration, Seattle, WA, United States

* Corresponding author: gaw@uw.edu

Table of Contents

1. Climate drivers with Rpath

BESTNPZ biomass outputs for euphausiids, copepods , microbes, and small and large phytoplankton are used to simulate climate-driven food web dynamics in the Rpath model. The functional groups whose biomass dynamics are represented by the BESTNPZ outputs include Euphausiids, Copepods, Pelagic microbes, Benthic microbes, Large phytoplankton, and Small phytoplankton. The Euphausiids, Large phytoplankton, and Small phytoplankton groups are forced with values from corresponding groups of similar name and taxonomic definition in the BESTNPZ output (Gibson and Spitz 2011, Kearney et al. 2020). Copepods are forced with the combined outputs from a large copepod group, which consists of *Neocalanus* sp. and *Calanus* sp., and a small coastal copepod group, which consists of smaller copepod taxa including, *Pseudocalanus* sp. Both the pelagic and benthic microbes groups are forced with the large microzooplankton group from the BESTNPZ. There is not an equivalent group in the BESTNPZ to our benthic microbes functional group that is available for forcing. So, both of our pelagic and benthic microbes are forced with the same outputs for microzooplankton. Because the units in the BESTNPZ output do not match the currency of our Rpath model, we forced it with monthly anomalies relative to the respective climatological monthly mean values over the hindcast period (1971–2017). Monthly anomalies over the forecast period (2018–2099) were calculated relative to the climatological monthly means over the hindcast period.

2. Bias-correction of climate drivers

Each earth system model is unique and presents a different image of the earth system. The underlying assumptions and uncertainty may differ between models, which can result in discrepancies among simulations (Ho et al. 2012). These systematic differences among models are particularly important when running simulations with multiple models or when joining a modeled historical period (i.e., hindcast) with a simulated forecast. The difference in variable values at the point where a hindcast ends and a simulation starts can be large and some form of correction for each model's unique predictions are necessary (Ho et al. 2012).

The BESTNPZ hindcast used in our Rpath simulations runs from 1991 through 2017, while the simulation forecasts with the RCPs begin in 2006. This gives us a reference period (overlap) of 12 years for the RCP model runs (2006–2017, Figure 1). We utilize this overlap to calculate a bias correction for the climate forcing variables during the projection period (2018–2099) in each simulated forecast that takes into account the mean and variability of the hindcast and the forecast during the reference period. We calculate bias-corrected trajectories for our forcing variables under each earth system model-RCP combination, following the method of Hawkins et al. (2013):

$$X_{BC,t} = \bar{X}_{hind,ref} + \frac{\sigma_{hind,ref}}{\sigma_{fut,ref}}(X_{fut,raw,t} - \bar{X}_{fut,ref}) \tag{1}$$

Where $X_{BC,t}$ is the bias-corrected (BC) variable projection at time-step *t* (months). $\bar{X}_{hind,ref}$ is the mean of the forcing variable from the hindcast (*hind*) during the reference period (*ref*). σ is the standard deviation of the variable during the reference period from either the hindcast (*hind*) or the projection (*fut*). $X_{fut,raw,t}$ is the raw uncorrected estimate of the forcing variable from the projections at time *t*. And $\bar{X}_{fut,ref}$ is the mean of the raw uncorrected forcing variable from the forecast during the reference period. This equation combines information from the hindcast and the forecast during a reference period to produce more realistic simulated futures (Hawkins et al. 2013). This method of bias correction is an additive bias correction (a.k.a., the "delta method") and can result in negative values. This was infrequently observed in our forcing data but when encountered, was addressed with a "zero trap" that did not permit our biological groups biomass to be ≤ 0.

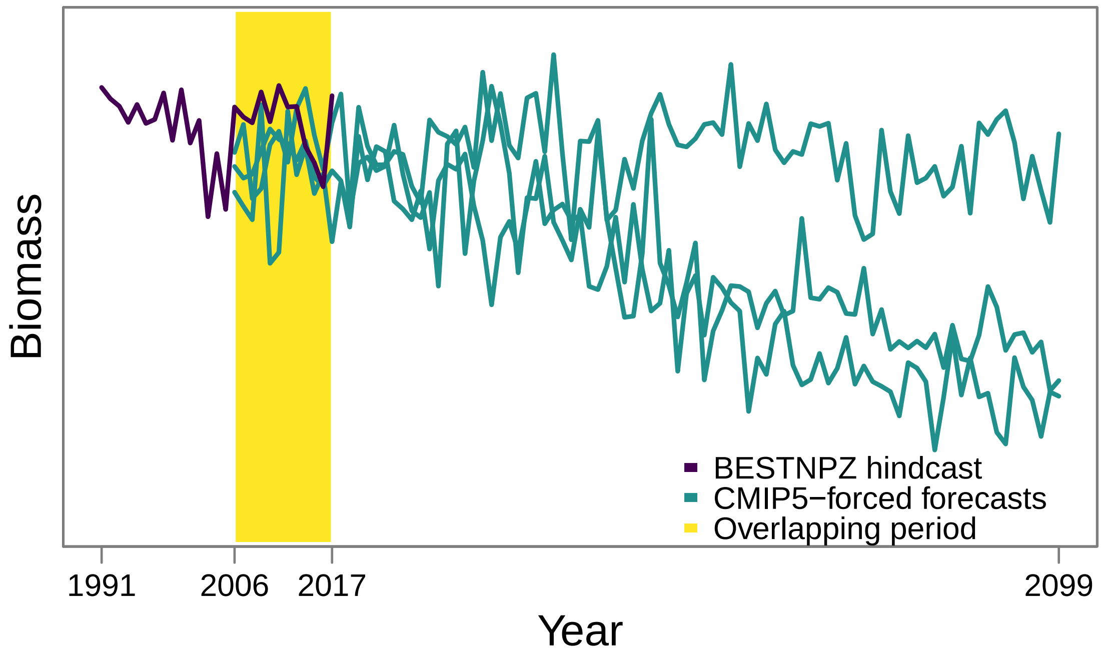

Figure 1: Schematic showing the overlapping period of time (2006–2017) in our simulations between the BESTNPZ hindcast (1991–2017) and the CMIP5-forced forecasts (2006–2099). The overlapping period is used to calculate bias-corrected trajectories for the projection period (2018–2099).

3. Functional group aggregation scheme

The Rpath model we assembled for this study is based on the eastern Bering Sea Ecopath with Ecosim model originally developed by Aydin et al. (2007). Their model was highly detailed with 129 biological groups, including 14 juvenile groups (a.k.a., stanzas). We maintained this level of functional group detail for many commercial and protected species that were of focal interest in this study. However, for groups that were data-poor or were not central interests to this study, we aggregated several species together into larger groups. Species were aggregated with other groups that had similar life history traits, food habits, and environmental requirements. Table 1 lists all the functional groups in the original model of Aydin et al. (2007) alongside the aggregated groups used in the present study.

Table 1: The aggregated functional groups used in our ACLIM Rpath model listed with the corresponding disaggregated groups from the full eastern Bering Sea model of Aydin et al. 2007.

Aggregated group	Group from Aydin et al. 2007
Transient killer whales	Transient killer whales
Toothed whales	Sperm and beaked whales
	Resident killer whales
	Porpoises
	Belugas
Gray whales	Gray whales
Baleen whales	Humpbacks
	Fin whales
	Sei whales
	Right whales
	Minke whales
	Bowhead whales
Pinnipeds	Sea otters
	Stellar sea lion juvenile
	Stellar sea lion adult
	Resident seals
Walrus and bearded seals	Walrus and bearded seals
Northern fur seal juvenile	Northern fur seal juvenile
Northern fur seal adult	Northern fur seal adult
Ice seals	Ice seals
Other birds	Shearwater
	Storm petrels
	Cormorants
	Gulls
Murres and puffins	Murres
	Puffins
Kittiwakes	Kittiwakes
Auklets	Auklets
Fulmars	Fulmars
Albatross	Albatross and Jaegers
Sharks	Sleeper shark
Walleye pollock juvenile	Walleye pollock juvenile
Walleye pollock adult	Walleye pollock adult
Pacific cod juvenile	Pacific cod juvenile
Pacific cod adult	Pacific cod adult
Herring	Herring juvenile
	Herring adult
Arrowtooth flounder juvenile	Arrowtooth flounder juvenile
Arrowtooth flounder adult	Arrowtooth flounder adult
Kamchatka flounder	Kamchatka flounder juvenile
	Kamchatka flounder adult
Greenland turbot juvenile	Greenland turbot juvenile
Greenland turbot adult	Greenland turbot adult
Pacific halibut juvenile	Pacific halibut juvenile
Pacific halibut adult	Pacific halibut adult
Yellowfin sole	Yellowfin sole juvenile
	Yellowfin sole adult
Flathead sole	Flathead sole juvenile
	Flathead sole adult
Northern rock sole	Northern rock sole juvenile
	Northern rock sole adult

Table 1 *continued*

Aggregated group	Group from Aydin et al. 2007
Alaska plaice	Alaska plaice
Other flatfish	Dover sole
	Rex sole
	Miscellaneous flatfish
Skates	Alaska skate
	Other skates
Sablefish	Sablefish juvenile
	Sablefish adult
Eelpouts	Eelpouts
Deep demersal fish	Grenadiers
	Miscellaneous deep fish
Pacific Ocean perch	Pacific Ocean perch
Other rock fish	Sharpchin rockfish
	Dusky rockfish
	Shortspine thornyhead
	Other *Sebastes*
Northern rockfish	Northern rockfish
Shortraker rockfish	Shortraker rockfish
Rougheye rockfish	Rougheye rockfish
Atka mackerel	Atka mackerel juvenile
	Atka mackerel adult
Shallow demersal fish	Greenlings
	Other sculpins
	Miscellaneous shallow fish
Large sculpins	Large sculpins
Octopi	Octopi
Squids	Squids
Salmon returning	Salmon returning
Salmon smolts	Salmon outgoing
Myctophids and Bathylagids	Bathylagidae
	Myctophidae
Capelin	Capelin
Sandlance	Sandlance
Other forage fish	Eulachon
	Other managed forage fish
	Other pelagic smelt
Tanner crab (*C. bairdi*)	Tanner crab (*C. bairdi*)
King crabs	King crabs
Snow crab (*C. opilio*)	Snow crab (*C. opilio*)
Pandalid shrimp	Pandalidae
Benthic zooplankton	Non-pandalid shrimp
	Miscellaneous crustaceans
	Benthic amphipods
	Mysids
Motile epifauna	Sea stars
	Brittle stars
	Urchins, dollars, cucumbers
	Snails
	Hermit crabs
	Miscellaneous crabs

Table 1 *continued*

Aggregated group	Group from Aydin et al. 2007
Structural epifauna	Anemones
	Corals
	Hydroids
	Urochordata
	Sea pens
	Sponges
Infauna	Bivalves
	Polychaetes
	Miscellaneous worms
Jellyfish	Scyphozoid jellies
	Gelatinous filter feeders
Pelagic zooplankton	Fish larvae
	Chaetognaths
	Pelagic amphipods
	Pteropods
Euphausiids	Euphausiids
Copepods	Copepods
Pelagic microbes	Pelagic microbes
Benthic microbes	Benthic microbes
Large phytoplankton	Macroalgae
	Large phytoplankton
	Outside production
Small phytoplankton	Small phytoplankton
Discards offal	Discards
	Offal
Pelagic detritus	Pelagic detritus
Benthic detritus	Benthic detritus
	Outside detritus
Trawl	Walleye pollock trawl
	Pacific cod trawl
	Atka mackerel trawl
	Northern rock sole trawl
	Yellowfin sole trawl
	Arrowtooth flounder trawl
	Flathead sole trawl
	Other flatfish trawl
	Greenland turbot trawl
	Rockfish trawl
Cod pots	Cod pots
Longline	Cod longline
	Greenland turbot longline
	Sablefish longline
Pacific halibut fishery	Halibut longline
Crab pots	Crab pots
Salmon fishery	Salmon fishery
Herring fishery	Herring fishery
Indigenous	Indigenous
Subsistence	Subsistence

4. Mass-balanced model parameters

Table 2: The parameters for our static, mass-balanced food web model of the eastern Bering Sea. TL is the trophic level calculated by Rpath. Biomass and Removals are in t km^{-2}. P/B (year^{-1}) is the production to biomass ratio, Q/B (year^{-1}) is the consumption to biomass ratio, and P/Q (year^{-1}) is the production to consumption ratio (also known as growth efficiency [GE]). EE is ecotrophic efficiency and is the proportion of production that is consumed by predators and removed by fisheries included in the model.

Functional group	TL	Biomass	P/B	Q/B	EE	P/Q	Removals
Transient killer whales	4.83	0.0001	0.03	11.16	0.00	0.002	0
Toothed whales	4.63	0.0349	0.07	17.42	0.03	0.004	3.19e-05
Gray whales	3.52	0.0327	0.06	8.87	0.04	0.007	0
Baleen whales	3.75	0.5079	0.03	6.67	0.09	0.004	1.93e-05
Pinnipeds	4.41	0.0157	0.09	20.78	0.08	0.004	6.93e-05
Walrus and bearded seals	3.56	0.1142	0.05	15.37	0.01	0.003	3.69e-05
Northern fur seal juvenile	4.59	0.0009	0.25	83.33	0.02	0.003	0
Northern fur seal adult	4.59	0.0326	0.09	39.03	0.04	0.002	3.70e-05
Ice seals	4.55	0.0303	0.07	19.20	0.87	0.004	3.19e-05
Other birds	4.52	0.0007	0.12	73.19	0.07	0.002	2.54e-06
Murres and puffins	4.42	0.0086	0.16	72.06	0.05	0.002	3.35e-05
Kittiwakes	4.40	0.0007	0.08	110.00	0.11	0.001	2.57e-06
Auklets	3.58	0.0018	0.17	110.00	0.05	0.002	6.81e-06
Fulmars	4.60	0.0005	0.06	73.00	0.15	0.001	2.02e-06
Albatross	4.61	0.0001	0.07	75.00	0.13	0.001	3.88e-07
Sharks	4.64	0.0532	0.10	3.00	0.16	0.033	0.0007
Walleye pollock juvenile	3.54	4.5669	1.45	8.50	0.81	0.171	0
Walleye pollock adult	3.68	18.4862	0.67	3.17	0.84	0.210	2.5091
Pacific cod juvenile	3.55	0.1868	1.78	8.88	0.79	0.201	0
Pacific cod adult	4.06	2.4650	0.41	2.28	0.51	0.181	0.4224
Herring	3.52	0.7827	0.48	4.45	0.68	0.108	0.0292
Arrowtooth flounder juvenile	3.80	0.0084	0.81	5.79	0.59	0.140	0
Arrowtooth flounder adult	4.31	0.9443	0.18	1.16	0.85	0.155	0.0361
Kamchatka flounder	4.45	0.0564	0.18	1.19	0.18	0.154	0
Greenland turbot juvenile	3.54	0.0075	1.70	7.91	0.99	0.215	0
Greenland turbot adult	4.59	0.3493	0.18	1.16	0.55	0.155	0.0090
Pacific halibut juvenile	3.52	0.0019	2.40	9.33	0.80	0.257	0
Pacific halibut adult	4.54	0.2219	0.19	1.10	0.69	0.173	0.0195
Yellowfin sole	3.56	5.2580	0.18	0.98	0.42	0.183	0.2361
Flathead sole	3.75	1.3412	0.27	2.21	0.30	0.124	0.0286
Northern rock sole	3.66	3.3830	0.23	1.25	0.45	0.188	0.1125
Alaska plaice	3.53	1.0684	0.20	2.00	0.53	0.100	0.0302
Other flatfish	3.68	0.2625	0.20	2.00	0.75	0.100	0.0079
Skates	4.18	0.7734	0.20	2.00	0.20	0.100	0.0283
Sablefish	4.28	0.0353	0.42	1.24	0.68	0.335	0.0028
Eelpouts	3.70	2.3715	0.40	2.00	0.80	0.200	0
Deep demersal fish	4.11	0.9762	0.15	2.00	0.27	0.075	0.0043

Table 2 *continued*

Functional group	TL	Biomass	P/B	Q/B	EE	GE	Removals
Pacific Ocean perch	3.54	0.1666	0.10	2.00	0.84	0.050	0.0103
Other rockfish	3.71	0.0179	0.11	1.61	0.63	0.070	0.0010
Northern rockfish	3.58	0.0279	0.10	2.00	0.80	0.050	0.0018
Shortraker rockfish	3.86	0.0095	0.10	2.00	0.45	0.050	0.0002
Rougheye rockfish	4.25	0.0037	0.10	2.00	0.61	0.050	0.0002
Atka mackerel	3.53	0.1440	0.75	7.29	0.25	0.103	0.0051
Shallow demersal fish	3.63	2.3056	0.40	2.00	0.80	0.200	4.71e-05
Large sculpins	3.96	0.5403	0.40	2.00	0.10	0.200	0.0100
Octopi	3.75	0.1925	0.80	3.65	0.80	0.219	0.0003
Squids	3.70	0.9270	3.20	10.67	0.80	0.300	0.0013
Salmon returning	3.82	0.1638	1.65	11.60	0.97	0.142	0.1698
Salmon smolts	3.55	0.0142	1.28	13.56	0.80	0.094	0
Myctophids and Bathylagids	3.52	0.9879	0.80	3.65	0.80	0.219	4.33e-07
Capelin	3.52	1.2447	0.80	3.65	0.80	0.219	1.28e-08
Sandlance	3.52	2.5277	0.80	3.65	0.80	0.219	4.48e-08
Other forage fish	3.52	2.1017	0.80	3.65	0.80	0.219	1.18e-04
Tanner crab (*C. bairdi*)	3.43	0.5219	1.60	3.09	0.71	0.518	0.0494
King crabs	3.51	0.2384	0.66	3.21	0.61	0.205	0.0196
Snow crab (*C. opilio*)	3.31	2.1788	1.30	3.12	0.51	0.415	0.3075
Pandalid shrimp	2.91	6.7425	0.58	2.41	0.80	0.239	2.21e-06
Benthic zooplankton	2.52	35.8847	4.90	23.86	0.80	0.206	0
Motile epifauna	2.82	9.8800	1.09	5.45	0.58	0.200	0.0073
Structural epifauna	2.53	0.7813	2.14	10.69	0.17	0.200	0.0026
Infauna	2.52	87.2299	1.75	8.77	0.31	0.200	1.35e-07
Jellyfish	2.60	1.1118	4.08	11.80	0.79	0.346	0.0186
Pelagic zooplankton	2.69	2.6433	3.57	10.19	0.81	0.350	0
Euphausiids	2.53	17.8310	5.48	15.64	0.80	0.350	0
Copepods	2.50	26.8574	6.00	27.74	0.80	0.216	0
Pelagic microbes	2	45.0000	36.50	104.29	0.26	0.350	0
Benthic microbes	2	22.0610	36.50	104.29	0.80	0.350	0
Large phytoplankton	1	9.1005	49.43		0.79		0
Small phytoplankton	1	39.5039	110.92		0.80		0

5. Large phytoplankton projections

Large phytoplankton are an important basal resource in the eastern Bering Sea food web, providing forage for zooplankton, ichthyoplankton, and input to detritus pools. The projections from Hermann et al. (2019) for our study region indicate a wide range of possible futures for large phytoplankton (Figure 2). The lower trophic level projections driven by the CESM and GFDL models indicate relatively minor increases in large phytoplankton biomass by the end of the century. However, the MIROC projection diverges from these other two projections with substantial increases for large phytoplankton.

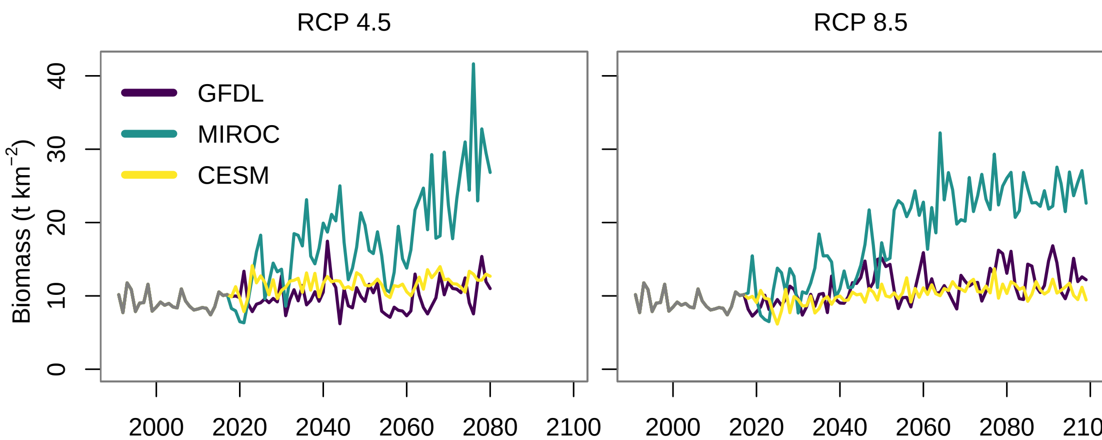

Figure 2: Biomass projections for large phytoplankton under all earth system model-RCP combinations. The gray line represents the hindcast period (1991–2017).

6. Sensitivity analysis

A mass-balanced Ecopath (or Rpath) model represents only one of many possible mass-balanced food web configurations. Additionally, there are varying degrees of uncertainty about all the model parameters, with some parameters being well supported by data and others less supported. Simulations with alternative model configurations can lead to divergent outcomes for species and the food web. We evaluated the sensitivity of our end-of-century simulation results using the Ecosense routine which generates an ensemble of Ecosim parameter sets from a single Ecopath model (Whitehouse and Aydin 2020). We conducted identical simulations with an ensemble of more than 2,000 unique parameterizations of our food web model to examine how sensitive our simulation results are to uncertainty in parameter estimates, including sensitivity to different parameterizations of the predator-prey functional response (i.e., vulnerability).

We subjected each of the generated Ecosim parameter sets to the same climate and fishery stressors as in the simulations in our main document text. The generated ecosystems were subjected to the observed fishery catch or specified fishing mortality rate during the hindcast period (1991–2017), and subjected to the fisheries sub-model (ATTACH) catch predictions during the projection period (2018–2099). In addition, we fixed the biomass trajectories of copepods, euphausiids, pelagic and benthic microbes, small phytoplankton, and large phytoplankton to be equal to their respective values from the reanalysis-forced BESTNPZ hindcast during the hindcast period. During the projection period, the biomass trajectories of these lower trophic level groups were fixed with their respective biomass trajectories from the CMIP5-forced BESTNPZ forecast using the GFDL earth system model under the RCP 8.5 emission scenario. Due to computing limitations and time constraints, it was not possible to repeat this sensitivity analysis with all earth system model-RCP combinations from the main text.

Full details of the Ecosense routine can be found in Whitehouse and Aydin (2020), but we provide a brief summary here. All food web model parameters were evaluated for quality using a data pedigree (Christensen et al. 2005), with specific definitions from Aydin et al. (2007). Each model parameter was assigned a pedigree that reflects the quality of the parameter estimate, in consideration of the data source, collection methodology, temporal and spatial coverage of the dataset, and taxonomic relevance. Each data pedigree corresponds to a prescribed range as a proportion of the point estimate (coefficient of variation, CV), intended to characterize parameter uncertainty, ranging from 0.1 to 0.8. Parameters that are well known and supported by an abundance of data, receive a good pedigree score (e.g., 1) which corresponds with a smaller CV (e.g., 0.1). Parameters with less support were assigned higher data pedigrees and thus higher CVs.

Full Ecosim parameter sets were drawn from distributions centered on the Ecopath parameter estimates and bounded by the specified CVs. Parameter estimates for biomass, P/B, Q/B, and other mortality (M_0) were drawn from a uniform distribution. The CV for a functional group's M_0 was set equal to the CV for P/B from the same functional group. Diet compositions are proportions and must sum to one and therefore were drawn from a Dirichlet distribution. The presence of a trophic link was not allowed to vary in the Ecosense routine, and the diet proportions drawn for all prey items present in the predator's Ecopath diet were not allowed to equal zero. We also varied the predator-prey functional response by varying the vulnerability parameter for each trophic link in the food web. Vulnerability in EwE ranges from one to infinity for each predator/prey trophic link, and is centered on 2.0; the value represents the ratio of top-down to bottom-up control plus 1.0 (so the center of 2.0 represents a balance of top-down versus bottom-up control). In practice, vulnerabilities greater than 91 are difficult to distinguish from infinity (Gaichas et al. 2012), so we vary vulnerability from 1.01 to 91 using a log-uniform distribution for all trophic trophic links. The pedigree scores and CVs for the original model parameters can be found in Aydin et al. (2007).

Each generated Ecosim parameter set was subjected to a 100-year burn-in period to eliminate numerically unstable configurations (Figure 3). During burn-in, if a functional group's biomass increased to more than one thousand times their starting biomass, or decreased to less than 1/1,000 their starting biomass, then the Ecosim parameter set was rejected and not retained for further analysis. Such ecosystem "crashes" generally occurred during the first few years of the burn-in period and were typically due to thermodynamically inconsistent parameter draws. Such as a parameter set drawn with insufficient prey production and exceptionally high predator consumption rates. The 100-year burn-in was generally sufficient to eliminate all such unstable food web configurations.

During burn-in, the generated ecosystems were also subjected to the observed 27 years of historical fishery catches (1991–2017) and the BESTNPZ hindcast biomass trajectories for copepods, euphausiids, pelagic and benthic microbes, small phytoplankton, and large phytoplankton. The observed catch and BESTNPZ hindcast trajectories were repeated as 27-year cycles during the 100-year burn-in period. This was done to eliminate Ecosim parameter sets during burn-in that would otherwise survive burn-in but crash once we began our projections with the forced fishery catches and forced lower trophic level biomass trajectories. More than 99.9% of the generated Ecosim parameter sets were rejected due to crashing during burn-in. We generated 2.55 million Ecosim parameter sets and retained 2,057 for our sensitivity analysis.

All of the retained Ecosim parameter sets were subjected to a simulation that was identical to the simulation in the manuscript main text with climate forcing under the GFDL earth system model and RCP 8.5 emission scenario, and subject to the status quo scenario of the fisheries sub-model. For each retained ecosystem, we calculated the percent change in biomass for each functional group as the change in the mean biomass from the final 10 years of the simulation relative to the mean of the respective group's biomass from the final 10 years of the hindcast period.

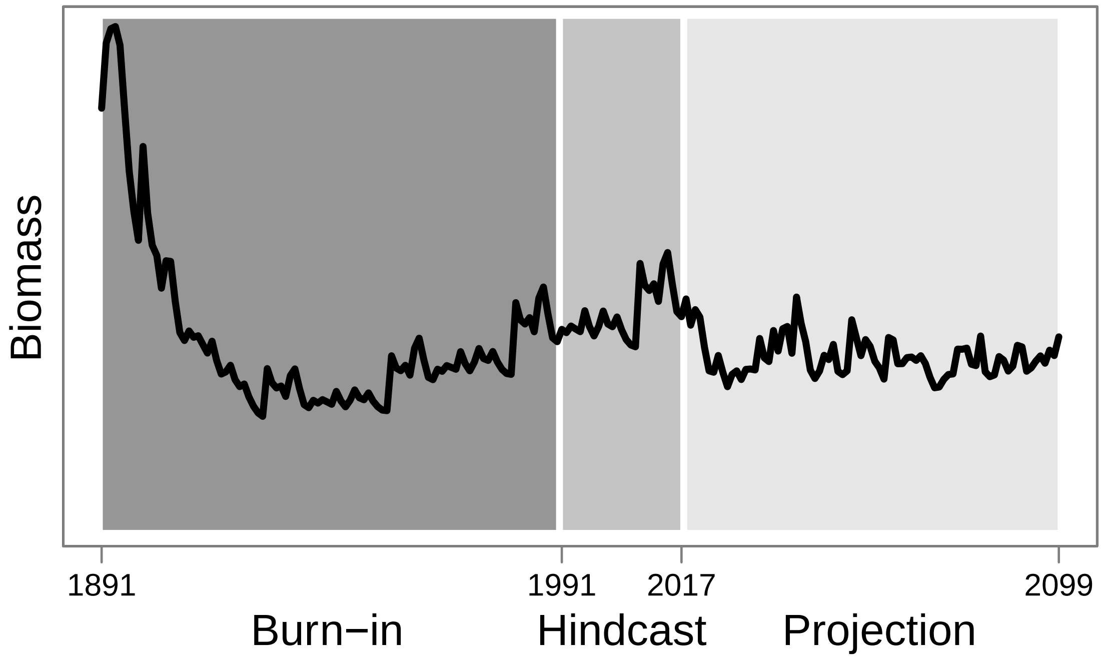

Figure 3: Schematic of the burn-in, hindcast, and projection periods for the sensitivity analysis simulations.

We calculated the percent change in biomass in the same fashion for the GFDL-RCP 8.5-status quo fishing scenario presented in the main document text, and compared that result with our sensitivity analysis in Figure 4. We found that for the majority of functional groups, the change in biomass from the simulation presented in our document main text (red dots in Figure 4) were in directional agreement with the median change in biomass for the same functional groups in our sensitivity analysis, and for 75% of the retained parameter sets. This indicates that the direction of change for most functional groups in our simulations are robust to parameter uncertainty, including uncertainty in the predator-prey functional response. For some groups, the boxplots were particularly wide and projections for those groups are less reliable (e.g., Pacific halibut juveniles, jellyfish, pelagic zooplankton).

Figure 4: (Figure on next page) Boxplots depicting the percent change in biomass for each functional group between the end of the hindcast period (2008–2017) and the end of the projection period (2090–2099), from each of the 2,057 Ecosim parameter sets in our sensitivity analysis. The black bars in the boxes represent the median percent change in biomass, the boxes are the interquartile range, the whiskers extend to the furthest point within 1.5x interquartile range, and outliers are not shown. The red dots represent the percent change biomass for each group from the model run with GFDL RCP 8.5 and the status quo fishing scenario from the main document text.

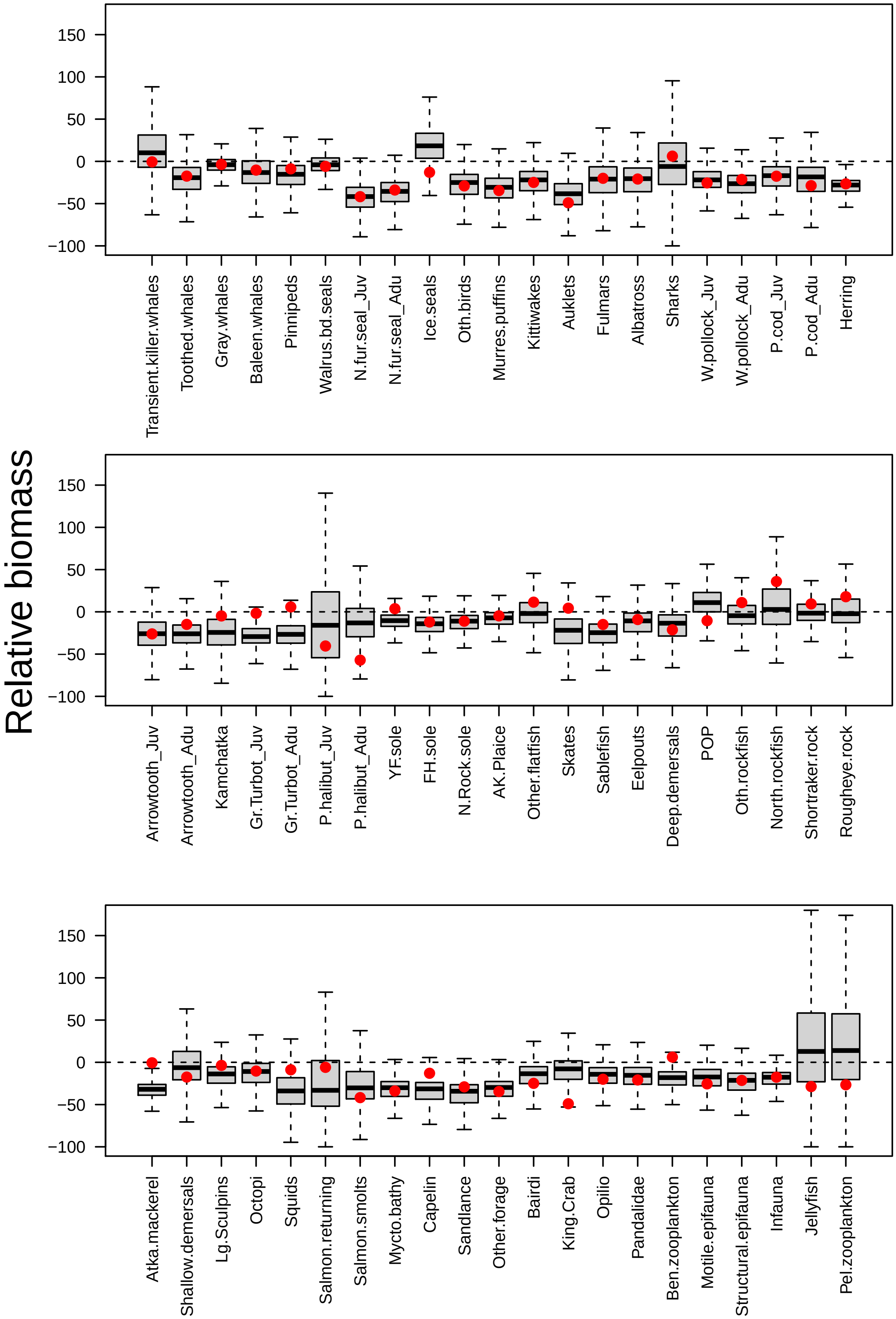

Figure 4: Caption on previous page.

7. Trophic guild biomass projections

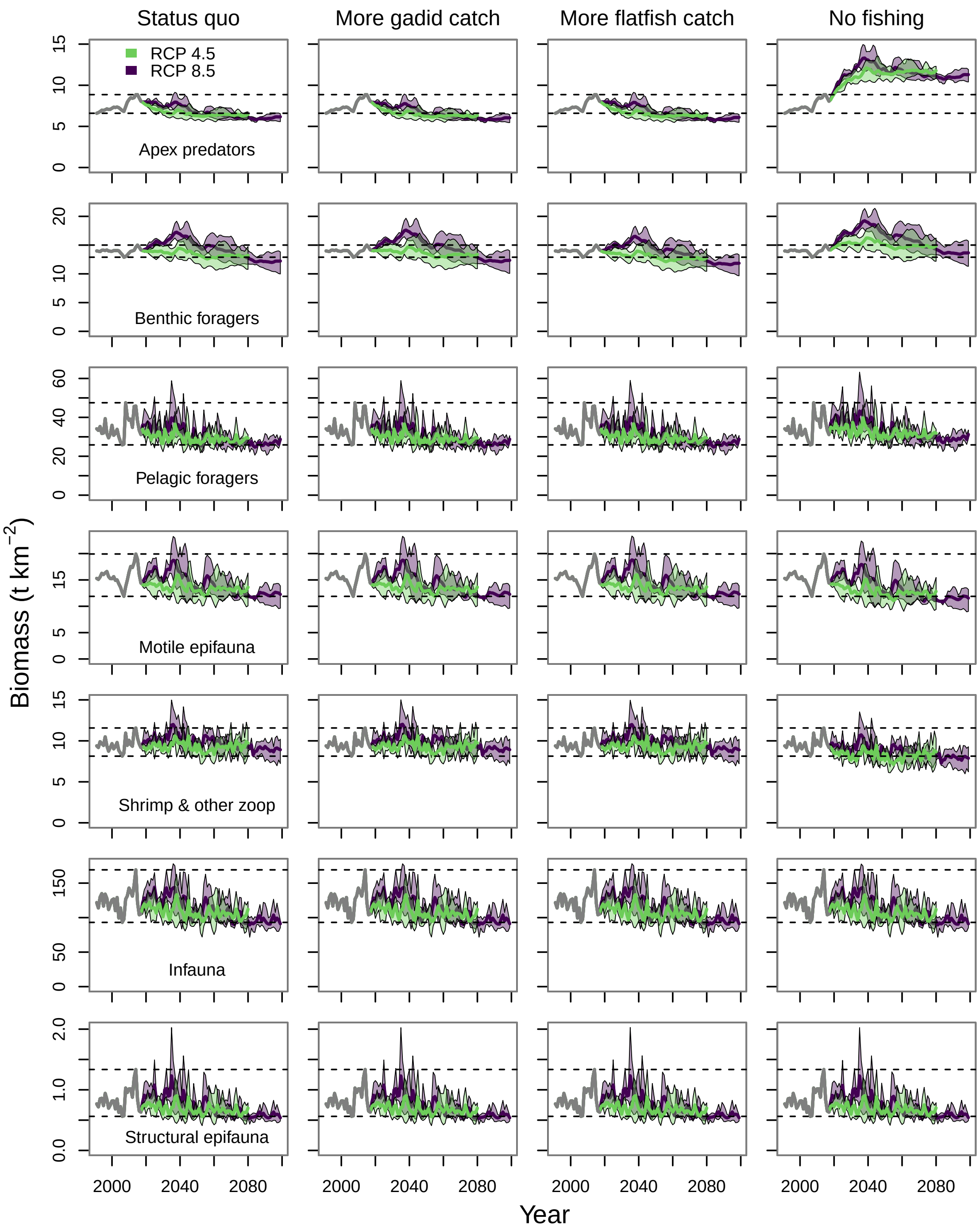

Figure 5: Trophic guild biomasses across the four fishing scenarios. The gray line from 1991–2017 indicates the hindcast period. The purple and green polygons indicate the minimum and maximum range for the three earth system models run under each RCP, purple for RCP 8.5 and green for RCP 4.5. The purple and green lines indicate the mean of the three earth system models for each RCP. And the dashed lines indicate the minimum and maximum values from the hindcast period.

8. Biomass projections of functional groups by trophic guild

The biomass projections for all functional groups in our food web model are in supplementary figures 4–10. The plots are organized by trophic guild, with all functional groups from a guild appearing together in multi-panel plots. All living functional groups are shown except for those lower trophic level groups whose biomass dynamics are shown in Figure 4 of the main document text.

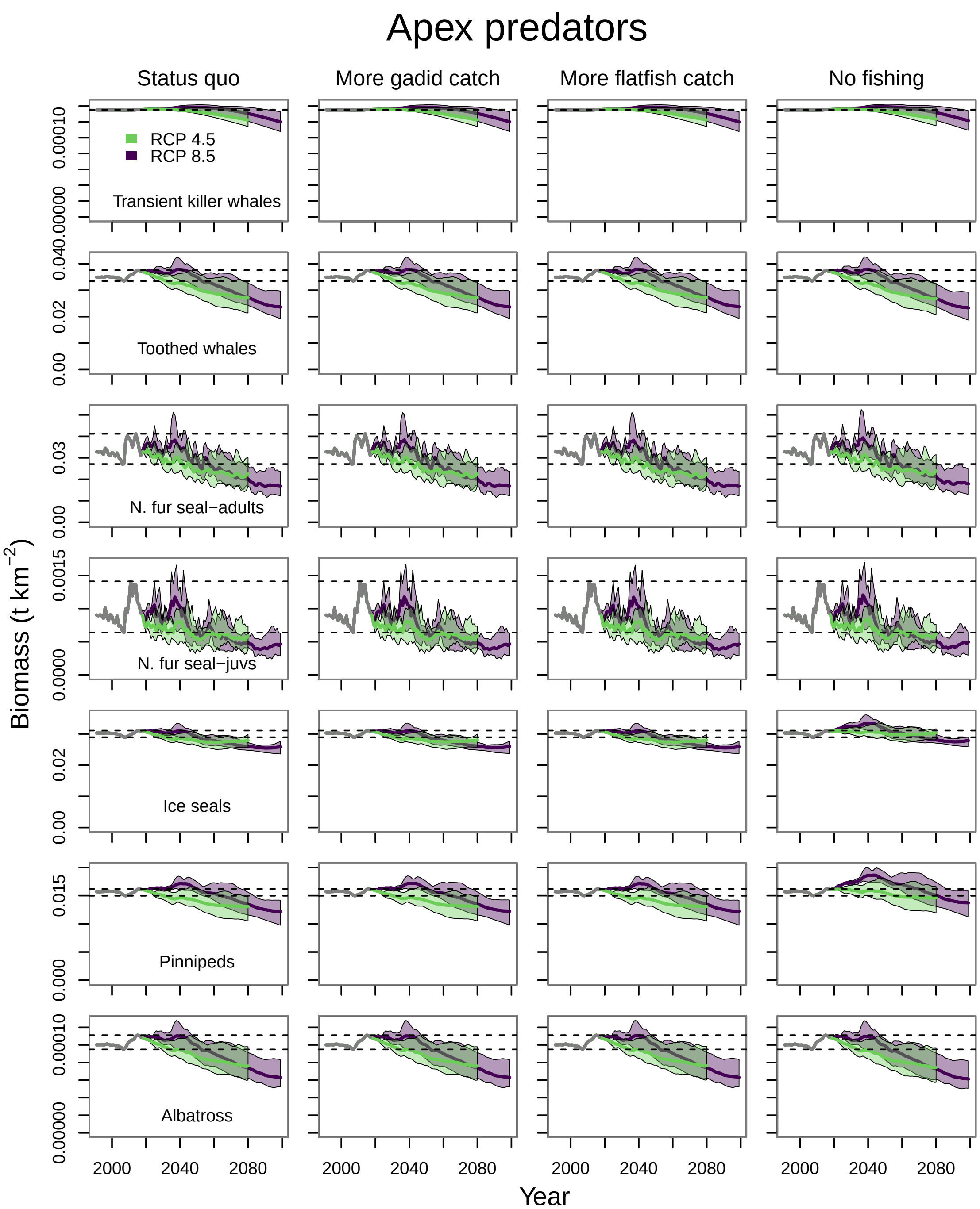

Figure 6: Biomass projections for functional groups from the apex predator guild across the four fishing scenarios. The gray line from 1991–2017 indicates the hindcast period. The purple and green polygons indicate the minimum and maximum range for the three ESMs run under each RCP, purple for RCP 8.5 and green for RCP 4.5. The purple and green lines indicate the mean of the three ESMs for each RCP. And the dashed lines indicate the minimum and maximum values from the hindcast period.

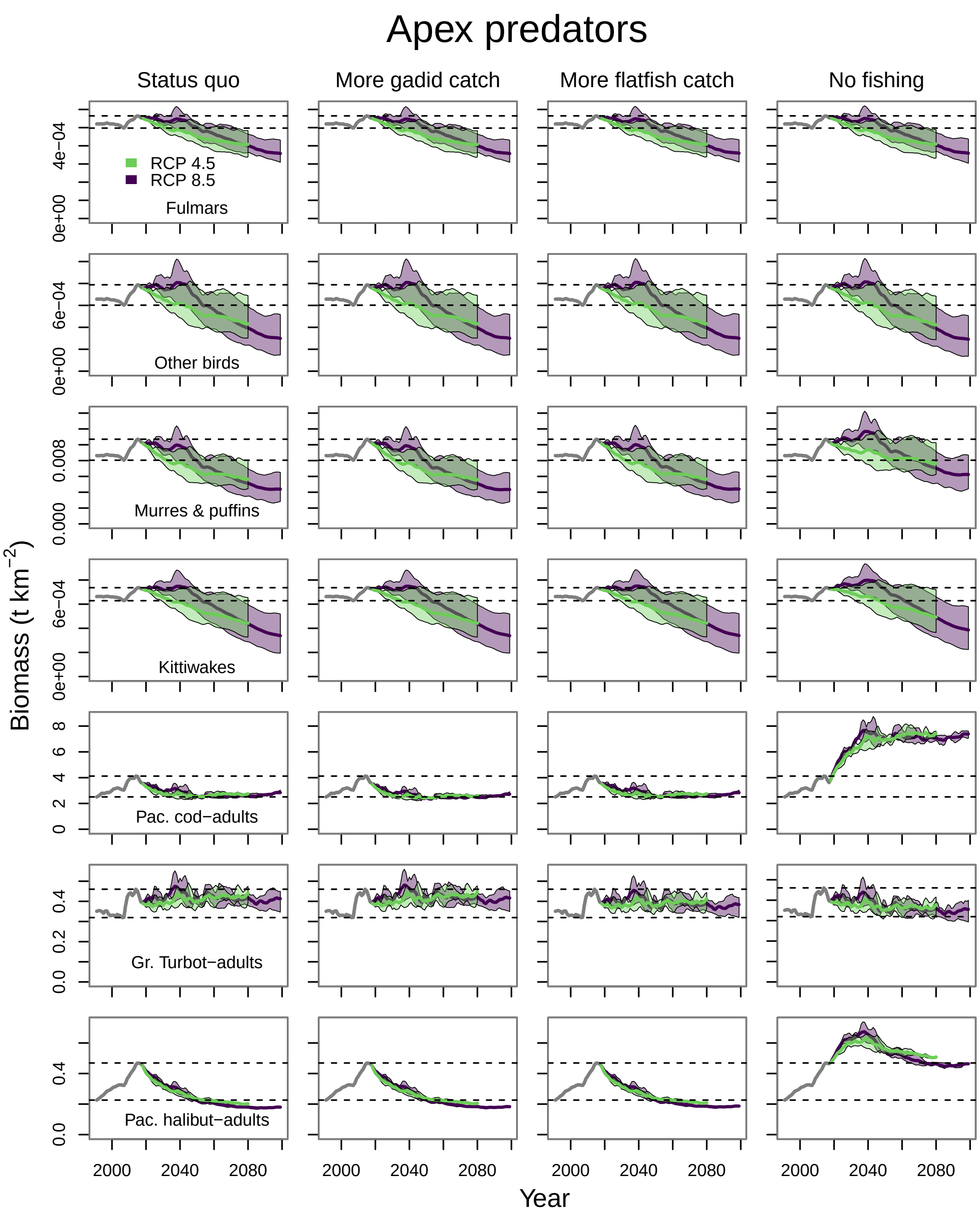

Figure 6: *continued*

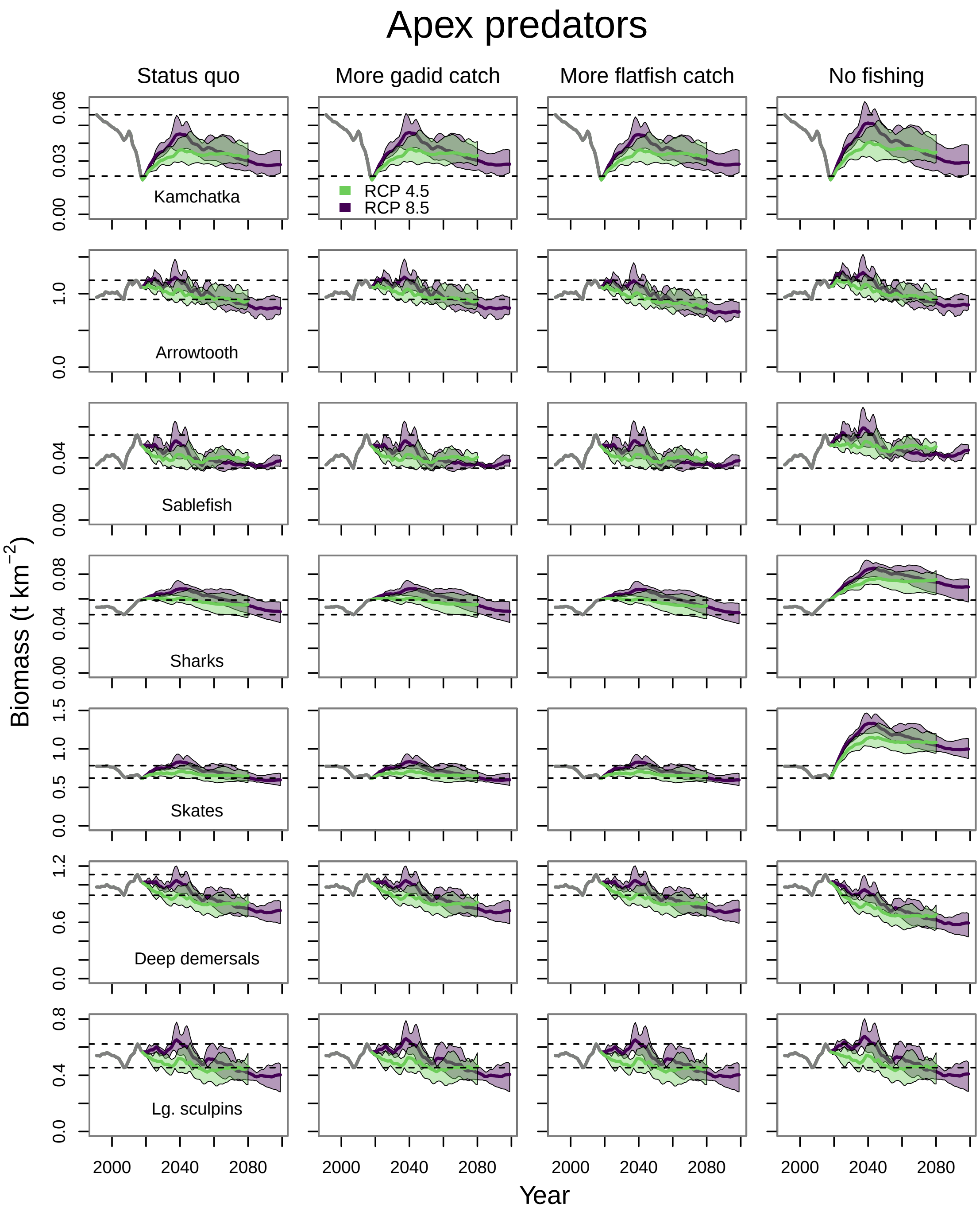

Figure 6: *continued*

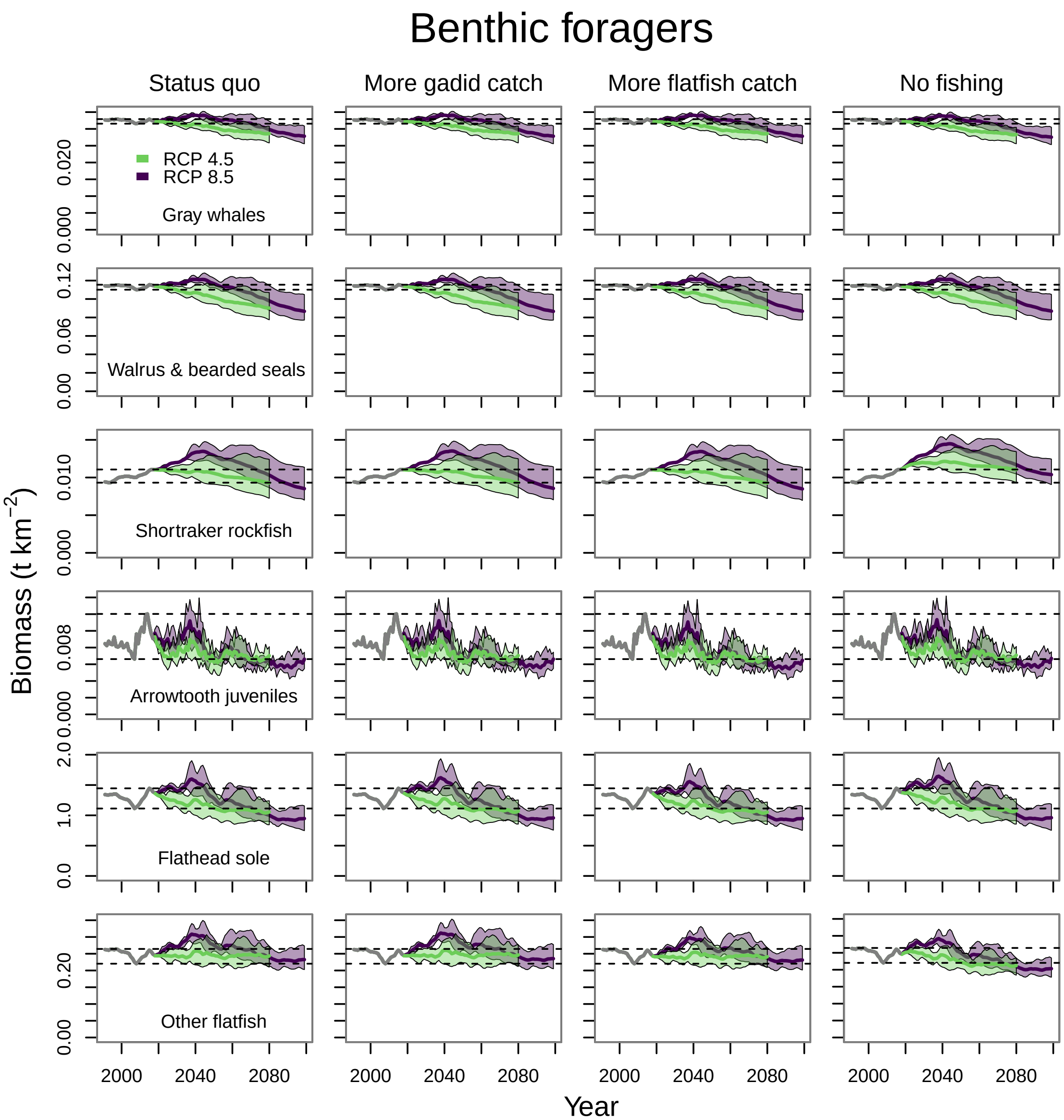

Figure 7: Biomass projections across the four fishing scenarios for functional groups from the benthic forager guild. The gray line from 1991–2017 indicates the hindcast period. The purple and green polygons indicate the minimum and maximum range for the three ESMs run under each RCP, purple for RCP 8.5 and green for RCP 4.5. The purple and green lines indicate the mean of the three ESMs for each RCP. And the dashed lines indicate the minimum and maximum values from the hindcast period.

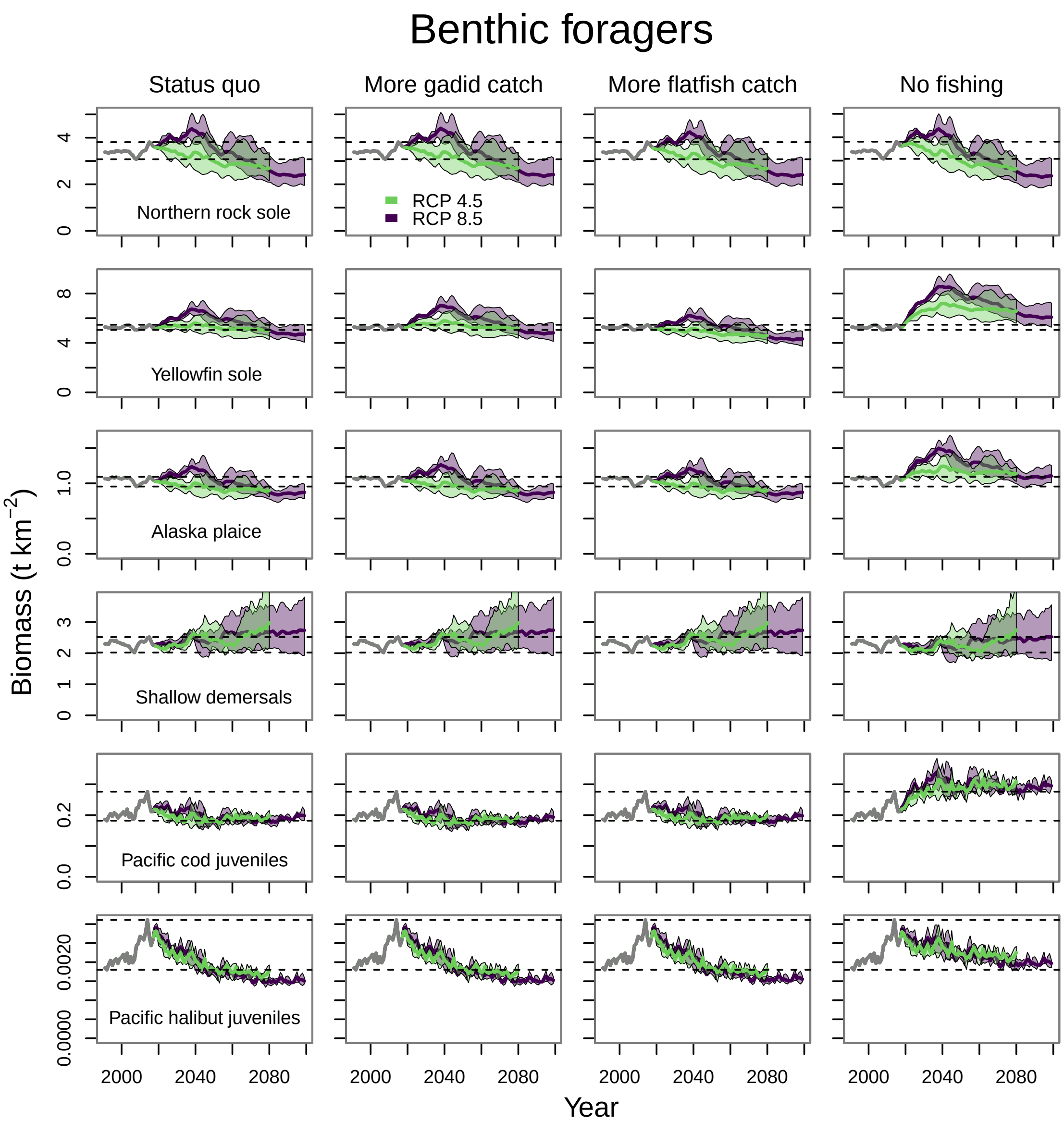

Figure 7: *continued*

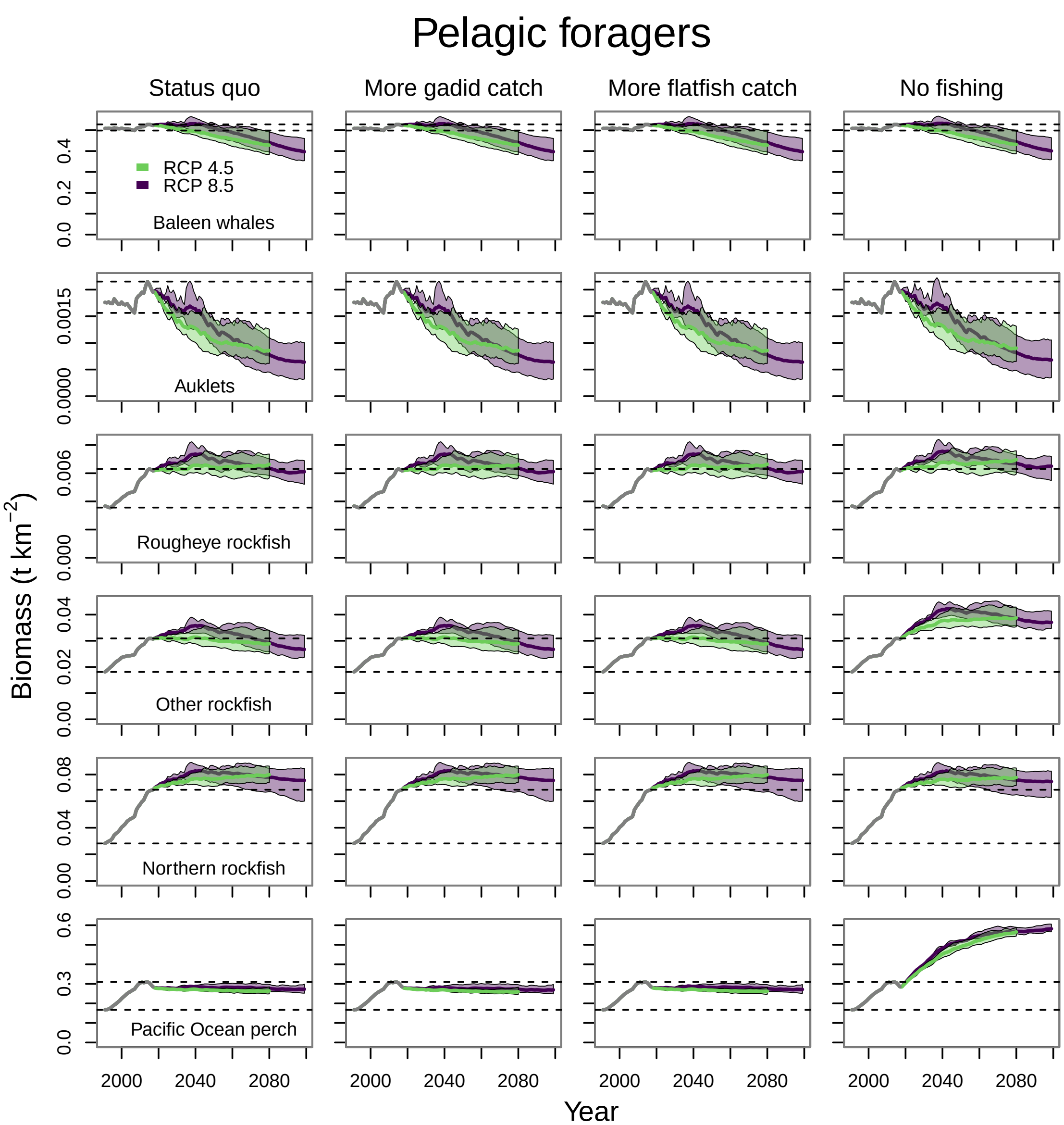

Figure 8: Biomass projections across the four fishing scenarios for functional groups from the pelagic forager guild. The gray line from 1991–2017 indicates the hindcast period. The purple and green polygons indicate the minimum and maximum range for the three ESMs run under each RCP, purple for RCP 8.5 and green for RCP 4.5. The purple and green lines indicate the mean of the three ESMs for each RCP. And the dashed lines indicate the minimum and maximum values from the hindcast period.

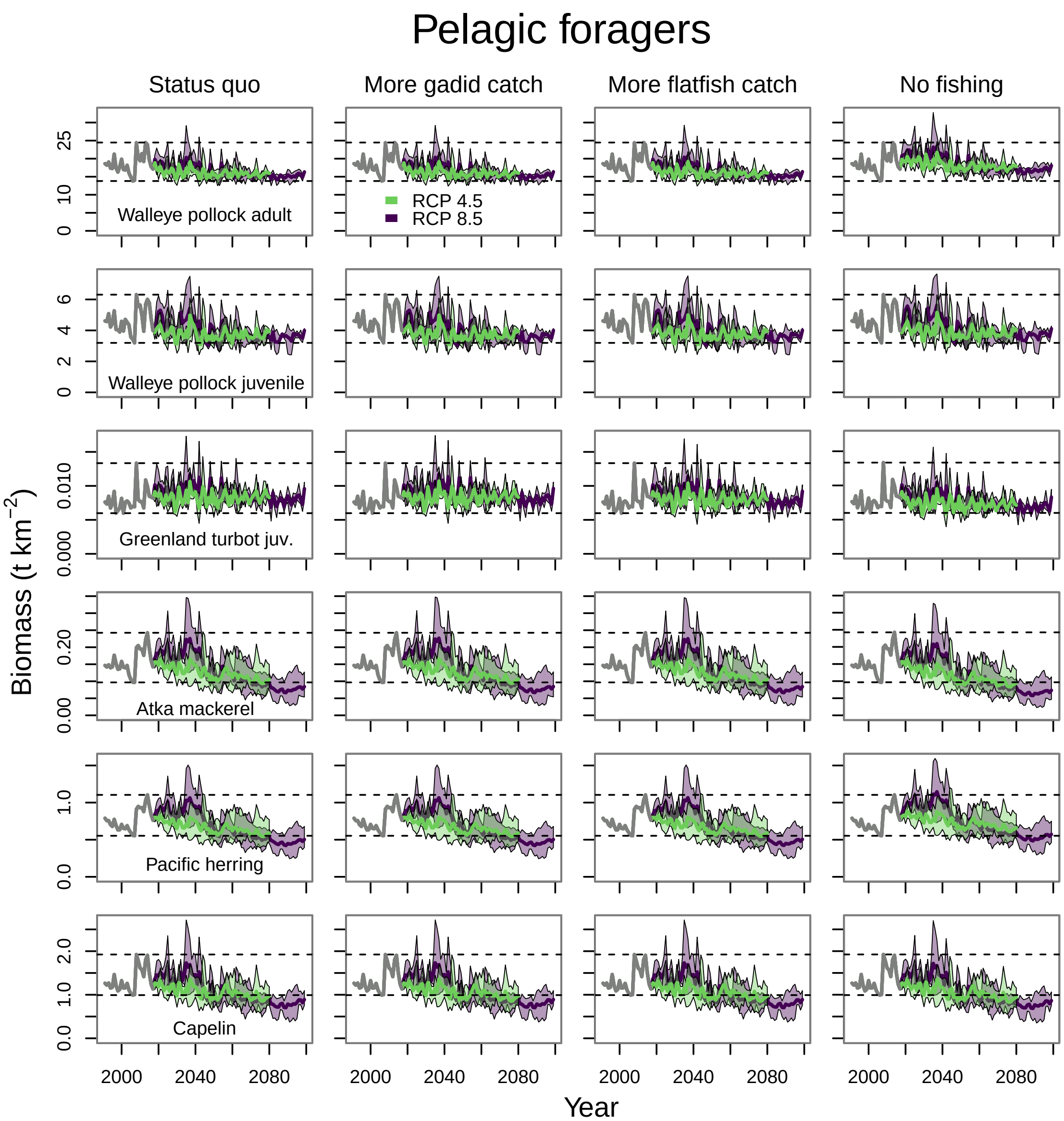

Figure 8: *continued*

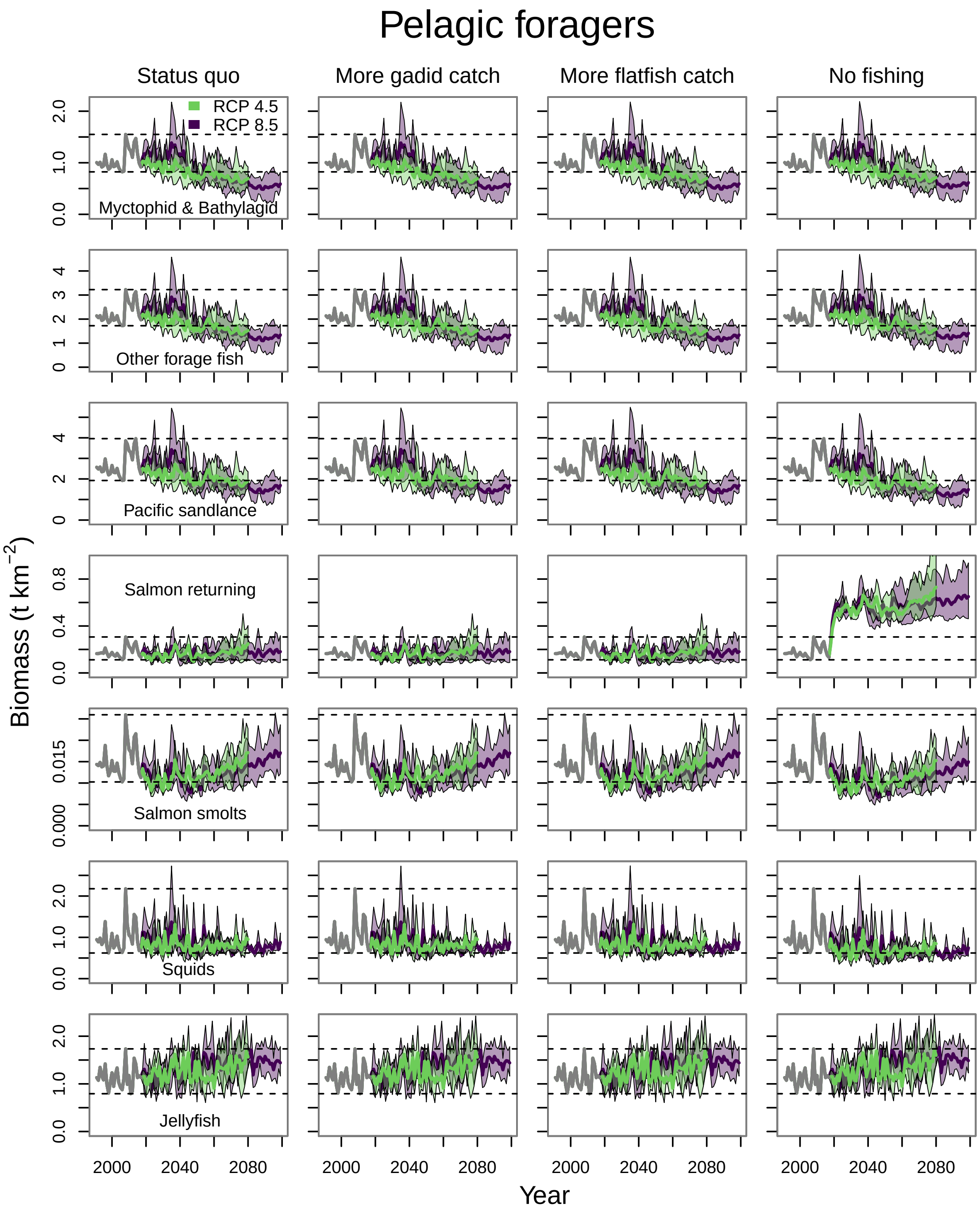

Figure 8: *continued*

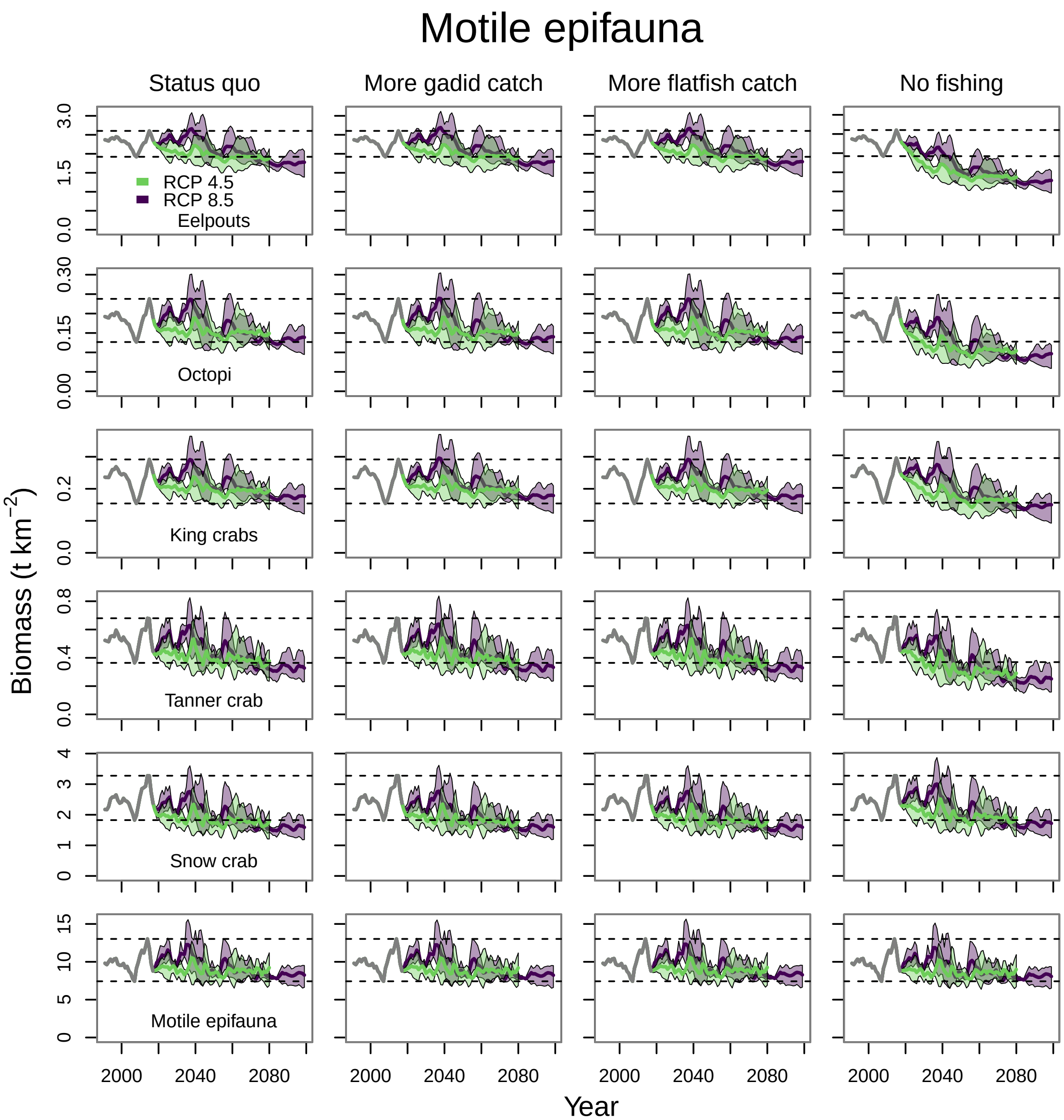

Figure 9: Biomass projections across the four fishing scenarios for functional groups from the motile epifauna guild. The gray line from 1991–2017 indicates the hindcast period. The purple and green polygons indicate the minimum and maximum range for the three ESMs run under each RCP, purple for RCP 8.5 and green for RCP 4.5. The purple and green lines indicate the mean of the three ESMs for each RCP. And the dashed lines indicate the minimum and maximum values from the hindcast period.

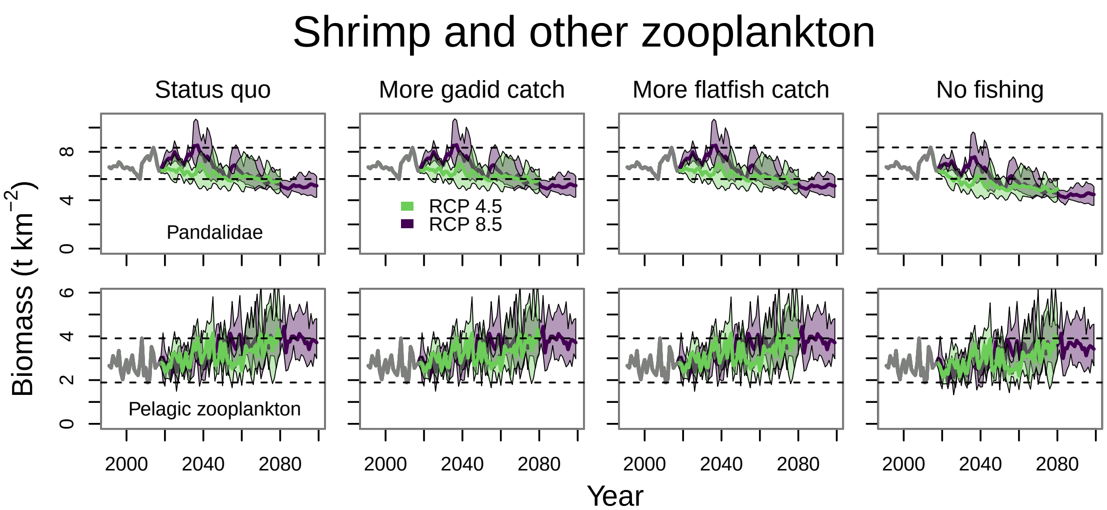

Figure 10: Biomass projections across the four fishing scenarios for functional groups from the shrimp and other zooplankton guild. The gray line from 1991–2017 indicates the hindcast period. The purple and green polygons indicate the minimum and maximum range for the three ESMs run under each RCP, purple for RCP 8.5 and green for RCP 4.5. The purple and green lines indicate the mean of the three ESMs for each RCP. And the dashed lines indicate the minimum and maximum values from the hindcast period.

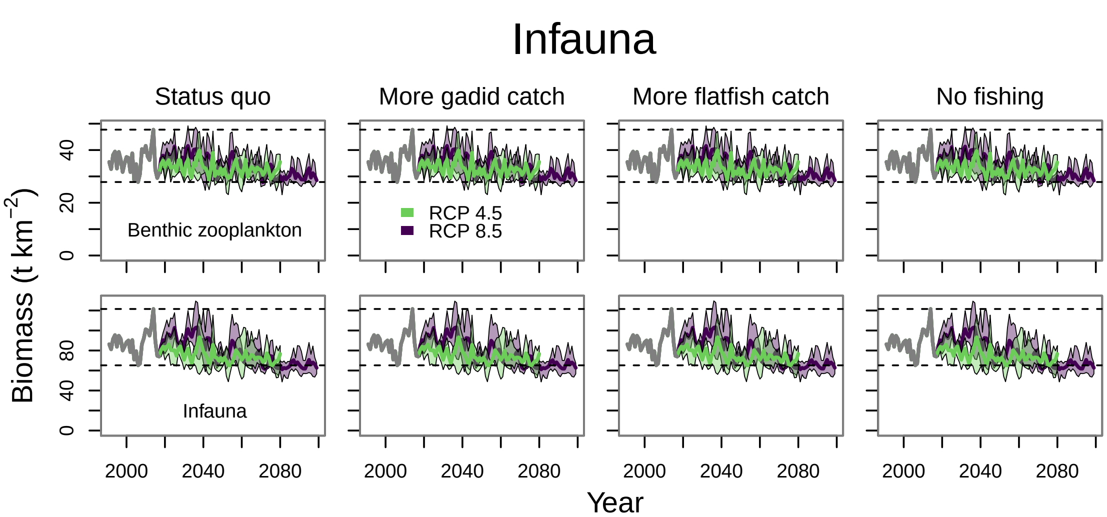

Figure 11: Biomass projections across the four fishing scenarios for functional groups from the infauna guild. The gray line from 1991–2017 indicates the hindcast period. The purple and green polygons indicate the minimum and maximum range for the three ESMs run under each RCP, purple for RCP 8.5 and green for RCP 4.5. The purple and green lines indicate the mean of the three ESMs for each RCP. And the dashed lines indicate the minimum and maximum values from the hindcast period.

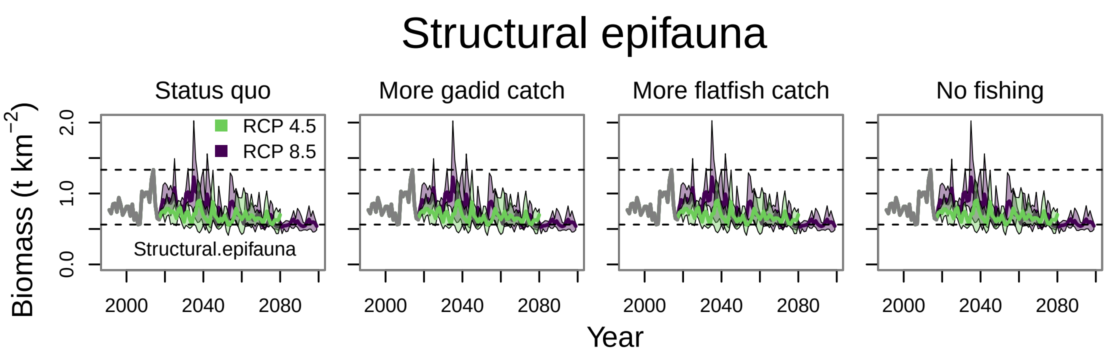

Figure 12: Biomass projections across the four fishing scenarios for functional groups from the structural epifauna guild. The gray line from 1991–2017 indicates the hindcast period. The purple and green polygons indicate the minimum and maximum range for the three ESMs run under each RCP, purple for RCP 8.5 and green for RCP 4.5. The purple and green lines indicate the mean of the three ESMs for each RCP. And the dashed lines indicate the minimum and maximum values from the hindcast period.

9. References

Aydin KY, Gaichas S, Ortiz I, Kinzey D, Friday N (2007) A comparison of the Bering Sea, Gulf of Alaska, and Aleutian Islands large marine ecosystems through food web modeling. U.S. Dep Commer, NOAA Tech. Memo. NMFS-AFSC-178.

Christensen V, Walters CJ, Pauly D (2005) Ecopath with Ecosim: A user's guide. Fisheries Centre, University of British Columbia, Vancouver, Canada, http://www.ecopath.org

Gaichas SK, Odell G, Aydin KY, Francis RC (2012) Beyond the defaults: functional response parameter space and ecosystem-level fishing thresholds in dynamic food web model simulations. Can J Fish Aquat Sci 69:2077-2094. doi:10.1139/f2012-099

Gibson GA, Spitz YH (2011) Impacts of biological parameterization, initial conditions, and environmental forcing on parameter sensitivity and uncertainty in a marine ecosystem model for the Bering Sea. J Mar Syst 88:214–231. doi:10.1016/j.jmarsys.2011.04.008

Hawkins E, Osborne TM, Ho CK, Challinor AJ (2013) Calibration and bias correction of climate projections for crop modelling: An idealised case study over Europe. Agric For Meteorol 170:19–31. doi:10.1016/j.agrformet.2012.04.007

Ho CK, Stephenson DB, Collins M, Ferro CAT, Brown SJ (2012) Calibration Strategies A Source of Additional Uncertainty in Climate Change Projections. Bull Amer Meteorol Soc 93:21–26. doi:10.1175/2011bams3110.1

Kearney K, Hermann A, Cheng W, Ortiz I, Aydin K (2020) A coupled pelagic-benthic-sympagic biogeochemical model for the Bering Sea: documentation and validation of the BESTNPZ model (v2019.08.23) within a high-resolution regional ocean model. Geosci Model Dev 13:597–650. doi:10.5194/gmd-13-597-2020

Whitehouse GA, Aydin KY (2020) Assessing the sensitivity of three Alaska marine food webs to perturbations: an example of Ecosim simulations using Rpath. Ecol Model 429. doi:10.1016/j.ecolmodel.2020.109074